

Database Schema

Management approach

Mechanism	Role
Hibernate <code>ddl-auto=update</code>	Creates/updates most tables from entities
<code>schema.sql</code>	Creates <code>account_ordinal_seq</code>
Seed runners	<code>RoleInitializer</code> , <code>AdminRoleInitializer</code> , <code>AccountPolicyInitializer</code>

Entity-relationship (current)

```
erDiagram
    customers ||--o{ account : has
    users ||--o{ user_roles : has
    roles ||--o{ user_roles : grants
    account_policy {
        bigint policy_id PK
        string account_type
        string currency_code
        boolean opening_deposit_required
        decimal required_opening_deposit
        decimal minimum_balance
        boolean active
    }
    customers {
        bigint customer_id PK
        string first_name
        string last_name
        string email UK
        string phone_number UK
        date date_of_birth
        string address
        string status
        timestamp created_at
        timestamp updated_at
    }
    account {
        bigint account_id PK
        string account_number UK
        string branch_code
        string account_type
        int ordinal
        string currency_code
        int check_digit
        decimal available_balance
        decimal ledger_balance
        int debit_count
        int credit_count
        string status
        bigint customer_id FK
    }
    users {
        bigint user_id PK
        string username UK
        string password_hash
        boolean enabled
        int failed_attempts
        boolean locked
    }
    roles {
        bigint role_id PK
        string name UK
    }
    user_roles {
        bigint user_role_id PK
        bigint user_id FK
        bigint role_id FK
    }
    account ||--o{ bank_transaction : posts
    bank_transaction {
        bigint transaction_id PK
        bigint account_id FK
    }
```

```

    string transaction_type
    decimal amount
    decimal balance_after
    string currency_code
    string narration
    timestamp created_at
    string created_by
}

```

Tables

customers

Entity: `com.bankone.customer.entity.Customer`

Column	Notes
<code>customer_id</code>	PK, identity
<code>first_name</code> , <code>last_name</code>	Required
<code>email</code> , <code>phone_number</code>	Unique
<code>date_of_birth</code>	Optional
<code>address</code> , <code>status</code>	Required
<code>created_at</code> , <code>updated_at</code>	@PrePersist / @PreUpdate

`customerCode` is **not** stored; JSON accessor formats ID via `BusinessIdFormatter`.

account

Entity: `com.bankone.account.entity.Account`

Column	Notes
<code>account_id</code>	PK
<code>account_number</code>	Unique, generated
<code>branch_code</code>	String (no branch table)
<code>account_type</code> , <code>currency_code</code> , <code>ordinal</code> , <code>check_digit</code>	Number composition
<code>available_balance</code> , <code>ledger_balance</code>	Updated on open/deposit
<code>debit_count</code> , <code>credit_count</code> , <code>last*_at</code>	Counters / stamps
<code>status</code>	ACTIVE, FROZEN, DORMANT, SUSPENDED, CLOSED
<code>customer_id</code>	FK → <code>customers</code>
Audit-ish	<code>created_at</code> , <code>activated_at</code> , <code>closed_at</code> , <code>created_by</code> , <code>closed_by</code>

account_policy

Entity: `com.bankone.account.entity.AccountPolicy` extends `AuditableEntity`

Unique `(account_type, currency_code)`.

Seeded INR policies for CURRENT, SAVINGS, SALARY, FIXED_DEPOSIT, RECURRING_DEPOSIT. **No LOAN seed.**

users , roles , user_roles

Entities under `com.bankone.user` / `com.bankone.role`. Sequences: `user_seq`, `role_seq`, `user_role_seq` (Hibernate-managed naming may vary by dialect config).

`AuditableEntity` columns on User/Role/UserRole/AccountPolicy: `created_at`, `updated_at`, `created_by`, `updated_by`, `version`.

bank_transaction

Entity: `com.bankone.transaction.entity.Transaction`

Table name is `bank_transaction` (avoids SQL keyword `transaction`).

Column	Notes
<code>transaction_id</code>	PK (identity)
<code>account_id</code>	FK → <code>account</code> (required)
<code>transaction_type</code>	Enum string: CREDIT, DEBIT
<code>amount</code>	> 0; precision 19,2
<code>balance_after</code>	Account ledger balance after posting
<code>currency_code</code>	Copied from account (length 3)
<code>narration</code>	Optional text
<code>created_at</code>	Set in <code>@PrePersist</code>
<code>created_by</code>	Staff username when provided

Written today from `AccountServiceImpl.deposit()` via `TransactionService.record` (CREDIT). No list API yet.

Sequences

```
-- schema.sql
CREATE SEQUENCE IF NOT EXISTS account_ordinal_seq
START WITH 1 INCREMENT BY 1;
```

Used by `AccountRepository.getNextOrdinal()`.

Not present (planned)

No tables yet for: `branch`, `loan` `product`, `audit_event`, `beneficiary`. Ledger table `bank_transaction` is present.

Migration notes

Changing entity fields with `ddl-auto=update` alters tables in place. For production-ready history, replace with Flyway/Liquibase (see `EXTENSION_GUIDE`).